

# Partial Derivatives and Gradients

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Outline (75 minutes)

1. **Functions of Several Variables** – Notation and examples (10 min)
2. **Partial Derivatives** – Definition, limit form, and computation (20 min)
3. **Higher-Order and Mixed Partial**s – Second derivatives, Clairaut's theorem (15 min)
4. **The Gradient** – Definition, geometry, and interpretation (15 min)
5. **Directional Derivatives** – Rates of change in any direction (10 min)
6. **Worked Examples and Practice** – Applications and exercises (5 min)

## 1 Functions of Several Variables

### 1.1 From One Variable to Many

*Recall:* A function of one variable  $f(x)$  assigns a number to each  $x$ . Its graph lives in 2D (the  $x$ - $y$  plane).

*Extension:* A function of **several variables**  $f(x_1, x_2, \dots, x_n)$  assigns a number to each  $n$ -tuple of inputs. We often write  $f(x, y)$  for two variables or  $f(x, y, z)$  for three.

### 1.2 Examples

*Example 1:*  $f(x, y) = x^2 + y^2$  — the squared distance from the origin in the plane.

*Example 2:*  $f(x, y) = xy$  — the area of a rectangle with sides  $x$  and  $y$ .

*Example 3:*  $f(x, y) = e^{-(x^2+y^2)}$  — a 2D Gaussian (bell surface).

*Example 4:*  $f(w_1, w_2, b) = \frac{1}{1+e^{-(w_1x_1+w_2x_2+b)}}$  — a sigmoid in two weights and a bias (neural network building block).

*Example 5:*  $f(w_1, w_2, \dots, w_n) = \sum_{i=1}^n w_i \mu_i - \lambda \sum_{i,j} w_i w_j \sigma_{ij}$  — portfolio expected return minus risk (finance).

## 2 Partial Derivatives

### 2.1 The Idea

When a function depends on several variables, we can ask: “How does  $f$  change if we vary *only*  $x$ , holding all other variables fixed?” That is the **partial derivative with respect to  $x$** .

### 2.2 Definition

**Definition 2.1.** The **partial derivative of  $f$  with respect to  $x$**  at  $(a, b)$  is the ordinary derivative of  $f(x, b)$  with respect to  $x$ , evaluated at  $x = a$ :

$$\frac{\partial f}{\partial x}(a, b) = \lim_{h \rightarrow 0} \frac{f(a + h, b) - f(a, b)}{h}$$

*Notation:* We write  $\frac{\partial f}{\partial x}$ ,  $f_x$ ,  $\partial_x f$ , or  $D_x f$  interchangeably. The symbol  $\partial$  (“partial”) reminds us that we vary only one variable.

*For  $n$  variables:* If  $f(x_1, \dots, x_n)$ , then

$$\frac{\partial f}{\partial x_i}(a_1, \dots, a_n) = \lim_{h \rightarrow 0} \frac{f(a_1, \dots, a_i + h, \dots, a_n) - f(a_1, \dots, a_i, \dots, a_n)}{h}$$

### 2.3 Computing Partial Derivatives

*Rule:* To find  $\frac{\partial f}{\partial x}$ , treat all other variables as **constants** and differentiate with respect to  $x$  using the usual rules (power rule, chain rule, etc.).

*Example:*  $f(x, y) = x^2y + 3xy^2$

- $\frac{\partial f}{\partial x} = 2xy + 3y^2$  (treat  $y$  as constant)
- $\frac{\partial f}{\partial y} = x^2 + 6xy$  (treat  $x$  as constant)

*Example:*  $f(x, y) = e^{x^2+y^2}$

- $\frac{\partial f}{\partial x} = e^{x^2+y^2} \cdot 2x = 2x e^{x^2+y^2}$  (chain rule: inner function  $x^2 + y^2$ , derivative of inner w.r.t.  $x$  is  $2x$ )
- $\frac{\partial f}{\partial y} = e^{x^2+y^2} \cdot 2y = 2y e^{x^2+y^2}$

Example:  $f(x, y) = \ln(x^2 + y^2)$

- $\frac{\partial f}{\partial x} = \frac{2x}{x^2 + y^2}$
- $\frac{\partial f}{\partial y} = \frac{2y}{x^2 + y^2}$

### 3 Higher-Order and Mixed Partial Derivatives

#### 3.1 Second Partial Derivatives

We can take partial derivatives of partial derivatives. For  $f(x, y)$ :

- $\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left( \frac{\partial f}{\partial x} \right) = f_{xx}$
- $\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left( \frac{\partial f}{\partial y} \right) = f_{yy}$
- $\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left( \frac{\partial f}{\partial x} \right) = f_{xy}$  — **mixed partial**: first  $x$ , then  $y$
- $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left( \frac{\partial f}{\partial y} \right) = f_{yx}$  — **mixed partial**: first  $y$ , then  $x$

#### 3.2 Clairaut's Theorem

**Theorem 3.1** (Clairaut's Theorem / Schwarz's Theorem). *If  $f_{xy}$  and  $f_{yx}$  are continuous on an open region, then*

$$f_{xy} = f_{yx}.$$

*The order of differentiation does not matter for mixed partials.*

Example: For  $f(x, y) = x^2y + 3xy^2$ :

$$\begin{aligned} f_x &= 2xy + 3y^2, & f_y &= x^2 + 6xy \\ f_{xx} &= 2y, & f_{yy} &= 6x \\ f_{xy} &= 2x + 6y, & f_{yx} &= 2x + 6y \quad \checkmark \end{aligned}$$

## 4 The Gradient

#### 4.1 Definition

**Definition 4.1.** The **gradient** of  $f(x_1, \dots, x_n)$  is the vector of all first partial derivatives:

$$\nabla f = \left( \frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right)$$

For two variables:  $\nabla f = \left( \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$ .

*Notation:*  $\nabla$  is read “nabla” or “grad.” Some texts write  $\text{grad } f$ .

#### 4.2 Geometric Interpretation

*Key fact:* The gradient  $\nabla f$  at a point points in the direction of **steepest ascent** (fastest increase of  $f$ ). Its magnitude  $|\nabla f|$  is the rate of increase in that direction.

*Level curves:* On a level curve  $f(x, y) = c$ , the gradient is **perpendicular** to the level curve, pointing toward higher values of  $f$ .

*Example:* For  $f(x, y) = x^2 + y^2$ , we have  $\nabla f = (2x, 2y)$ . At  $(1, 1)$ ,  $\nabla f = (2, 2)$  points away from the origin (where  $f$  is smallest). The level curves are circles; the gradient is radial (perpendicular to the circles).

#### 4.3 Why the Gradient Points Uphill

If we take a small step  $\mathbf{h}$  from a point, the change in  $f$  is approximately

$$f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) \approx \nabla f \cdot \mathbf{h} = |\nabla f| |\mathbf{h}| \cos \theta$$

where  $\theta$  is the angle between  $\nabla f$  and  $\mathbf{h}$ . This is maximized when  $\cos \theta = 1$ , i.e., when  $\mathbf{h}$  points in the direction of  $\nabla f$ . So the gradient is the direction of steepest ascent.

### 5 Directional Derivatives

#### 5.1 Definition

**Definition 5.1.** The **directional derivative** of  $f$  at  $\mathbf{x}$  in the direction of a unit vector  $\mathbf{u}$  is

$$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} = \frac{\partial f}{\partial x}u_1 + \frac{\partial f}{\partial y}u_2 + \cdots$$

*Interpretation:*  $D_{\mathbf{u}}f$  is the rate of change of  $f$  as we move in direction  $\mathbf{u}$ .

*Special cases:*

- If  $\mathbf{u} = (1, 0)$ , then  $D_{\mathbf{u}}f = \frac{\partial f}{\partial x}$ .
- If  $\mathbf{u} = (0, 1)$ , then  $D_{\mathbf{u}}f = \frac{\partial f}{\partial y}$ .
- The maximum directional derivative is  $|\nabla f|$ , achieved when  $\mathbf{u}$  points in the direction of  $\nabla f$ .

### 5.2 Example

For  $f(x, y) = x^2 + y^2$  at  $(1, 1)$ ,  $\nabla f = (2, 2)$ . In the direction  $\mathbf{u} = \frac{1}{\sqrt{2}}(1, 1)$ :

$$D_{\mathbf{u}}f = (2, 2) \cdot \frac{1}{\sqrt{2}}(1, 1) = \frac{4}{\sqrt{2}} = 2\sqrt{2} = |\nabla f|$$

So moving in the gradient direction gives the maximum rate of increase.

## 6 Worked Examples

### 6.1 Example 1: Quadratic Form

Let  $f(x, y) = x^2 - 2xy + 3y^2 + x - y$ .

Partials:

$$\frac{\partial f}{\partial x} = 2x - 2y + 1, \quad \frac{\partial f}{\partial y} = -2x + 6y - 1$$

Gradient:

$$\nabla f = (2x - 2y + 1, -2x + 6y - 1)$$

At  $(1, 1)$ :  $\nabla f(1, 1) = (1, 3)$ . The steepest ascent from  $(1, 1)$  is in the direction  $(1, 3)$  (or the unit vector  $\frac{1}{\sqrt{10}}(1, 3)$ ).

### 6.2 Example 2: Gaussian

Let  $f(x, y) = e^{-(x^2+y^2)}$ .

Partials:

$$\frac{\partial f}{\partial x} = -2x e^{-(x^2+y^2)}, \quad \frac{\partial f}{\partial y} = -2y e^{-(x^2+y^2)}$$

Gradient:

$$\nabla f = -2e^{-(x^2+y^2)}(x, y)$$

At the origin  $(0, 0)$ ,  $\nabla f = (0, 0)$  — a critical point (the peak of the bell).

### 6.3 Example 3: Cobb-Douglas Production

In economics,  $f(L, K) = L^\alpha K^{1-\alpha}$  models output from labor  $L$  and capital  $K$  ( $0 < \alpha < 1$ ).

Partials:

$$\frac{\partial f}{\partial L} = \alpha L^{\alpha-1} K^{1-\alpha} = \alpha \frac{f}{L}, \quad \frac{\partial f}{\partial K} = (1-\alpha) L^\alpha K^{-\alpha} = (1-\alpha) \frac{f}{K}$$

Gradient:

$$\nabla f = \left( \alpha \frac{f}{L}, (1 - \alpha) \frac{f}{K} \right)$$

## 7 Summary

### 7.1 Key Formulas

$$\begin{aligned} \frac{\partial f}{\partial x} &= \text{treat } y \text{ constant, differentiate w.r.t. } x \\ \nabla f &= \left( \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) \\ D_{\mathbf{u}} f &= \nabla f \cdot \mathbf{u} \quad (\text{unit vector } \mathbf{u}) \end{aligned}$$

### 7.2 Practice Exercises

*Exercise 1:* For  $f(x, y) = x^3y^2 + \sin(xy)$ , find  $f_x$ ,  $f_y$ , and  $\nabla f$  at  $(1, 0)$ .

*Exercise 2:* For  $f(x, y) = \ln(x^2 + 2y^2)$ , compute  $f_{xx}$ ,  $f_{yy}$ , and  $f_{xy}$ .  
Verify  $f_{xy} = f_{yx}$ .

*Exercise 3:* For  $f(x, y) = x^2 + 4xy + y^2$ , find the direction of steepest ascent at  $(2, -1)$  and the rate of change in that direction.

*Exercise 4:* A temperature field is  $T(x, y) = 100e^{-(x^2+y^2)/4}$ . At the point  $(1, 1)$ , in which direction does temperature increase fastest? What is that rate?

*Exercise 5:* Let  $f(x, y, z) = x^2 + yz - z^2$ . Compute  $\nabla f$  and evaluate it at  $(1, 2, 1)$ .

### Worked Solutions

*Exercise 1 (Solution):*  $f(x, y) = x^3y^2 + \sin(xy)$ .

$$f_x = 3x^2y^2 + y \cos(xy), \quad f_y = 2x^3y + x \cos(xy)$$

At  $(1, 0)$ :  $f_x(1, 0) = 0 + 0 = 0$ ,  $f_y(1, 0) = 0 + 1 \cdot 1 = 1$ . So  $\nabla f(1, 0) = (0, 1)$ .  $\square$

*Exercise 2 (Solution):*  $f(x, y) = \ln(x^2 + 2y^2)$ . Then

$$f_x = \frac{2x}{x^2 + 2y^2}, \quad f_y = \frac{4y}{x^2 + 2y^2}$$

$$f_{xx} = \frac{2(x^2 + 2y^2) - 2x \cdot 2x}{(x^2 + 2y^2)^2} = \frac{2x^2 + 4y^2 - 4x^2}{(x^2 + 2y^2)^2} = \frac{4y^2 - 2x^2}{(x^2 + 2y^2)^2}$$

$$f_{yy} = \frac{4(x^2 + 2y^2) - 4y \cdot 4y}{(x^2 + 2y^2)^2} = \frac{4x^2 + 8y^2 - 16y^2}{(x^2 + 2y^2)^2} = \frac{4x^2 - 8y^2}{(x^2 + 2y^2)^2}$$

$$f_{xy} = \frac{-2x \cdot 4y}{(x^2 + 2y^2)^2} = \frac{-8xy}{(x^2 + 2y^2)^2}, \quad f_{yx} = \frac{-4y \cdot 2x}{(x^2 + 2y^2)^2} = \frac{-8xy}{(x^2 + 2y^2)^2}$$

So  $f_{xy} = f_{yx}$ .  $\square$

*Exercise 3 (Solution):*  $f(x, y) = x^2 + 4xy + y^2$ . Then  $\nabla f = (2x + 4y, 4x + 2y)$ . At  $(2, -1)$ :  $\nabla f = (0, 6)$ . The direction of steepest ascent is  $(0, 6)$  or the unit vector  $(0, 1)$ . The rate is  $|\nabla f| = 6$ .  $\square$

*Exercise 4 (Solution):*  $T(x, y) = 100e^{-(x^2+y^2)/4}$ . Then

$$\nabla T = 100e^{-(x^2+y^2)/4} \cdot \left(-\frac{x}{2}, -\frac{y}{2}\right) = -50e^{-(x^2+y^2)/4}(x, y)$$

At  $(1, 1)$ :  $\nabla T = -50e^{-1/2}(1, 1) \approx -30.3(1, 1)$ . Temperature increases fastest in the direction opposite to  $\nabla T$ , i.e., toward the origin:  $\mathbf{u} = \frac{1}{\sqrt{2}}(1, 1)$  or  $(-1, -1)$  normalized. The rate is  $|\nabla T| \approx 30.3\sqrt{2} \approx 42.9$  degrees per unit distance.  $\square$

*Exercise 5 (Solution):*  $f(x, y, z) = x^2 + yz - z^2$ . Then

$$\nabla f = (2x, z, y - 2z)$$

At  $(1, 2, 1)$ :  $\nabla f = (2, 1, 0)$ .  $\square$